2

Quantum statistical mechanics

We now start what will be the subject of this book: quantum field theory at non-zero temperature and/or non-zero chemical potential. Two competing formalisms have been used at zero temperature in order to study field theory: the operator formalism, which is the older one, and the path integral formalism, which represents the more modern approach. Actually, it may often be illuminating to look at a given problem from both points of view, although in some cases one of the formalisms may prove to be definitely superior to the other: for example, path integrals are much simpler when quantizing gauge theories. At finite temperature, both formalisms are useful, and it is instructive to be able to switch from one approach to the other. Since the simplest case of field theory is field theory with zero space dimension, or, in other words, quantum mechanics, we shall begin with a short description of quantum mechanics at finite temperature, or, equivalently, quantum statistical mechanics. We first deal with the path integral approach, and later on revert to the more conventional operator formalism. We shall be particularly interested in time-ordered products of position operators, which will generalize to time-ordered products of field operators in field theory.

At zero temperature, it is often convenient to perform an analytical continuation from real to imaginary time: $t \to -i\tau$, or $x^0 \to -ix_4$, where τ (x_4) is real. This also means going from Minkowski to Euclidean space, since the Minkowski metric transforms into a Euclidean metric (with a change of sign):

$$t^2 - \mathbf{x}^2 \to -(\tau^2 + \mathbf{x}^2) \tag{2.1}$$

In momentum space the corresponding operation is $k^0 \to -ik_4$. For reasons which will become clear in the following chapters, the use of Euclidean space plays a much more important role at finite temperature than at zero temperature. Thus the first section of the present chapter will

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be devoted to a short review of the path integral formalism in imaginary time.

2.1 Path integral formalism and imaginary time

In conventional quantum mechanics, the motion of a particle in a time-independent potential $V(q)$ can be described by the probability amplitude $F(q',t';q,t)$ of finding the particle at position q' at time t', when one knows that it was located at point q at time t:

$$F(q',t';q,t) =< q'|e^{-i\hat{H}(t'-t)}|q > \tag{2.2}$$

$\hat{H}$ is the full, time-independent Hamiltonian; for simplicity, we restrict ourselves to a one-dimensional motion, and we set $\hbar = 1$. Whenever there is a risk of confusion, circumflexes will be used to denote operators acting in the Hilbert space of states. There exists a path integral representation of F, which can be found in textbooks and will not be described here, as we shall be interested in the analytical continuation of F to imaginary time:

$$\begin{gathered} t \to -i\tau \qquad t' \to -i\tau' \\ F(q',-i\tau';q,-i\tau) =< q'|e^{-\hat{H}(\tau'-\tau)}|q > \end{gathered} \tag{2.3}$$

Let us derive a path integral representation for F in (2.3). We proceed in a heuristic way and ignore the subtleties linked to the definition of the integration measure. We divide the interval $[\tau,\tau']$ into $(n+1)$ subintervals, each of length $\varepsilon = (\tau - \tau')/(n+1)$, with $n \to \infty$, and write

$$\exp[-\hat{H}(\tau' - \tau)] = \exp\left[-(\tau' - \tau)\left(\frac{\hat{p}^2}{2m} + V(\hat{q})\right)\right] \tag{2.4}$$

where $\hat{q}$ and $\hat{p}$ are the position and momentum operators. We then use Trotter's (or Lie product) formula:

$$\lim_{n\to\infty}\left(e^{A/n}e^{B/n}\right)^n = e^{A+B} \tag{2.5}$$

Equation (2.5) is easily proved when A and B are bounded operators, and the necessary generalization to unbounded operators can be found in the literature: see, for example, Glimm and Jaffe (1987). We insert complete sets of eigenstates of the position operator $\hat{q}$ at 'times' $\tau_1, ..., \tau_n$

$$\begin{aligned} F(q',-i\tau';q,-i\tau) = \lim_{\varepsilon\to 0}\int \prod_{l=1}^{n} dq_l < q_{l+1}|\exp\left(-\frac{\varepsilon}{2}V(\hat{q})\right) \\ \times \exp\left(-\varepsilon\frac{\hat{p}^2}{2m}\right)\exp\left(-\frac{\varepsilon}{2}V(\hat{q})\right)|q_l > \end{aligned} \tag{2.6}$$

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and evaluate the matrix elements in (2.6). The action of $V(\hat{q})$ on the ket $|q_l>$ is of course trivial, while for the matrix element of the momentum operator we insert a complete set of states $|p_l>$ of the momentum operator $\hat{p}$:

$$\begin{aligned} &< q_{l+1}|\exp\left(-\varepsilon\frac{\hat{p}^2}{2m}\right)|q_l> \\ &\quad = \int \mathrm{d}p_l < q_{l+1}|\exp\left(-\varepsilon\frac{\hat{p}^2}{2m}\right)|p_l >< p_l|q_l > \\ &\quad = \int \frac{\mathrm{d}p_l}{2\pi} \mathrm{e}^{i(q_{l+1}-q_l)p_l} \mathrm{e}^{-\varepsilon p_l^2/2m} = \left(\frac{m}{2\pi\varepsilon}\right)^{1/2} \mathrm{e}^{-m(q_{l+1}-q_l)^2/2\varepsilon} \end{aligned} \tag{2.7}$$

Equation (2.7) allows us to rewrite F as

$$\begin{aligned} F(q', -i\tau'; q, -i\tau) = \lim_{\varepsilon\to 0}\left(\frac{m}{2\pi\varepsilon}\right)^{1/2} \int \prod_{l=1}^{n} \left[\left(\frac{m}{2\pi\varepsilon}\right)^{1/2} \mathrm{d}q_l\right] \\ \times \exp\left[-\varepsilon\left(\sum_{l=0}^{n} \frac{m(q_{l+1}-q_l)^2}{2\varepsilon^2}\right.\right. \\ \left.\left. + \sum_{l=0}^{n} V\left(\frac{q_l + q_{l+1}}{2}\right)\right)\right] \end{aligned} \tag{2.8}$$

We have written the argument of V as $(q_l + q_{l+1})/2$ for aesthetic reasons and have neglected some terms on the boundary, which is possible in the limit $\varepsilon \to 0$. Proceeding in a formal way, we call $\mathscr{D}q(\tau'')$ the limit $\varepsilon \to 0$ of the integration measure in (2.8) and recognize in the argument of the exponential a Riemann sum for the integral:

$$S_E(\tau' - \tau) = \int_\tau^{\tau'} \mathrm{d}\tau'' \left[\frac{1}{2} m\dot{q}^2(\tau'') + V(q(\tau''))\right] \tag{2.9}$$

where the dot indicates a derivative with respect to τ; S_E in (2.9) is often called the Euclidean action. We may finally write F in the form of a path integral:

$$\begin{aligned} F(q', -i\tau'; q, -i\tau) = \int \mathscr{D}q(\tau'') \\ \times \exp\left[-\int_\tau^{\tau'} \mathrm{d}\tau'' \left(\frac{1}{2} m\dot{q}^2(\tau'') + V(q(\tau''))\right)\right] \end{aligned} \tag{2.10}$$

The boundary conditions on the paths $q(\tau'')$ are $q(\tau) = q$, $q(\tau') = q'$. Although the above 'derivation' is purely formal, it should be emphasized that a rigorous meaning can be given to the integration measure in (2.10). Standard derivations of (2.10) use the Hamiltonian form of the path integral as an intermediate step (see exercise 2.1). Here we have chosen a more direct derivation, which is valid because the Hamiltonian has the

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form given in (2.4). In more complicated cases, it may be necessary to go through the Hamiltonian form of the path integral.

At this point we make the connection with quantum statistical mechanics by using (2.10) to express the partition function $Z(\beta)$ as a path integral; as usual $\beta = 1/T$ is the inverse of the temperature in a system of units where the Boltzmann constant $k_B = 1$. We start from the definition of the partition function:

$$Z(\beta) = \mathrm{Tr}\ \mathrm{e}^{-\beta\hat{H}} = \sum_n \mathrm{e}^{-\beta E_n} \tag{2.11}$$

where the trace has been evaluated by using a complete set of eigenvectors of $\hat{H}$:

$$\hat{H}|n> = E_n|n> \tag{2.12}$$

However, it is also possible to write the trace by using a complete set of eigenvectors of the position operator:

$$Z(\beta) = \int \mathrm{d}q < q|\mathrm{e}^{-\beta\hat{H}}|q > \tag{2.13}$$

It is clear that $\mathrm{e}^{-\beta\hat{H}}$ can be interpreted formally as an evolution operator in imaginary time. Comparing with (2.3) we see that

$$Z(\beta) = \int \mathrm{d}q F(q, -i\beta; q, 0) \tag{2.14}$$

so that $Z(\beta)$ can be expressed as a path integral:

$$\begin{aligned} Z(\beta) &= \int \mathscr{D}q(\tau) \exp\left[-\int_0^\beta \mathrm{d}\tau \left(\frac{1}{2}m\dot{q}^2(\tau) + V(q(\tau))\right)\right] \\ &= \int \mathscr{D}q(\tau) \exp\left[-S_E(\beta)\right] \end{aligned} \tag{2.15}$$

where one integrates over paths subject to the boundary condition

$$q(\beta) = q(0) \tag{2.16}$$

namely, over paths with period β in imaginary time.

As usual in quantum field theory, we define a generating functional $Z(\beta; j)$, with $Z(\beta) = Z(\beta; j = 0)$, through

$$Z(\beta; j) = \int \mathscr{D}q(\tau) \exp\left[-S_E(\beta) + \int_0^\beta j(\tau)q(\tau)\mathrm{d}\tau\right] \tag{2.17}$$

Functional differentiation of (2.17) gives the propagator in imaginary time:

$$\frac{1}{Z(\beta)} \frac{\delta^2 Z(\beta; j)}{\delta j(\tau_1)\delta j(\tau_2)}\bigg|_{j=0} = \frac{1}{Z(\beta)} \int \mathscr{D}q(\tau) q(\tau_1) q(\tau_2) \mathrm{e}^{-S_E(\beta)} \tag{2.18}$$

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Indeed, we now show that the RHS of (2.18) can be identified as the thermal average of a time-ordered-, or T-product of position operators:

$$\langle T(\hat{q}(-i\tau_1)\hat{q}(-i\tau_2))\rangle_\beta = \frac{1}{Z(\beta)}\mathrm{Tr}\left[\mathrm{e}^{-\beta\hat{H}}T(\hat{q}(-i\tau_1)\hat{q}(-i\tau_2))\right] \tag{2.19}$$

where $\hat{q}(-i\tau)$ is the position operator in the Heisenberg picture:

$$\begin{aligned}\hat{q}(t) &= \mathrm{e}^{i\hat{H}t}\hat{q}\ \mathrm{e}^{-i\hat{H}t}\\ \hat{q}(-i\tau) &= \mathrm{e}^{\hat{H}\tau}\hat{q}\ \mathrm{e}^{-\hat{H}\tau}\end{aligned} \tag{2.20}$$

We recall that the thermal average $\langle\hat{A}\rangle_\beta$ of an operator $\hat{A}$ is defined by

$$\langle\hat{A}\rangle_\beta = \frac{1}{Z(\beta)}\mathrm{Tr}\left(\hat{A}\mathrm{e}^{-\beta\hat{H}}\right) \tag{2.21}$$

The T-product in (2.19) operates in imaginary time:

$$\begin{aligned}T(\hat{q}(-i\tau_1)\hat{q}(-i\tau_2)) &= \hat{q}(-i\tau_1)\hat{q}(-i\tau_2) \text{ if } \tau_1 > \tau_2\\ &= \hat{q}(-i\tau_2)\hat{q}(-i\tau_1) \text{ if } \tau_2 > \tau_1\end{aligned} \tag{2.22}$$

That the functional differentiation of $Z(\beta;j)$ does give the thermal average of a T-product of position operators is easily proved: one writes, for example, for $\tau_1 > \tau_2$

$$\begin{aligned}&Z(\beta)\langle T(\hat{q}(-i\tau_1)\hat{q}(-i\tau_2))\rangle_\beta\\ &\quad = \int \mathrm{d}q < q|\mathrm{e}^{-(\beta-\tau_1)\hat{H}}\hat{q}\mathrm{e}^{-(\tau_1-\tau_2)\hat{H}}\hat{q}\mathrm{e}^{-\tau_2\hat{H}}|q >\end{aligned} \tag{2.23}$$

Then one inserts a complete set of states of the position operator at 'times' τ_1 and τ_2 and repeats the procedure leading to the path integral (2.15). It is also straightforward to convince oneself that $Z(\beta;j)$ can be written in operator form:

$$Z(\beta;j) = \mathrm{Tr}\left[\mathrm{e}^{-\beta\hat{H}}T\left(\mathrm{e}^{\int_0^\beta \mathrm{d}\tau j(\tau)\hat{q}(-i\tau)}\right)\right] \tag{2.24}$$

From the cyclicity of the trace in the operator form of the thermal average, or from the periodicity (2.16) of the paths in the path integral formalism, we deduce the following property of the propagator:

$$\langle T(\hat{q}(-i\beta)\,\hat{q}(-i\tau))\rangle_\beta = \langle T(\hat{q}(0)\hat{q}(-i\tau))\rangle_\beta \tag{2.25}$$

Until now we have restricted τ to lie in the interval $[0,\beta]$; we shall see in the next section that $\Delta(\tau)$ is defined *a priori* in the interval $[-\beta,\beta]$. Then the cyclicity of the trace allows us to define a function $\Delta(\tau)$ which is periodic in imaginary time:

$$\Delta(\tau) = \langle T(\hat{q}(-i\tau)\hat{q}(0))\rangle_\beta \tag{2.26}$$

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and which obeys

$$\Delta(\tau - \beta) = \Delta(\tau) \tag{2.27}$$

for any value of τ in the interval $[0, \beta]$.

Up to now we have not specified the potential $V(q)$; we now specialize to the quantum mechanical analogue of the free field, namely, the harmonic oscillator with potential

$$V(q) = \frac{1}{2}\omega^2 q^2 \tag{2.28}$$

In order to simplify the formulae and to make a smooth connection with the following chapters, we set $m = 1$. We wish to compute the time-ordered product of position operators (2.19), which will generalize to the free field propagator. We use the path integral approach: the results will be rederived later on using the operator formalism. The generating functional $Z(\beta; j)$ is easily computed, since the integration is now Gaussian after an integration by parts in $S_E(\beta)$:

$$\begin{aligned} Z(\beta; j) &= \int_{q(0)=q(\beta)} \mathcal{D}q(\tau) \exp\left[-\int_0^\beta d\tau \left(\frac{1}{2}q(\tau)\right.\right. \\ &\quad \left.\left. \times \left(-\frac{d^2}{d\tau^2} + \omega^2\right)q(\tau) - j(\tau)q(\tau)\right)\right] \\ &= Z(\beta) \exp\left[\frac{1}{2}\int d\tau d\tau' j(\tau) K(\tau, \tau') j(\tau')\right] \end{aligned} \tag{2.29}$$

In (2.29), the Green function $K(\tau, \tau')$ is the inverse of $(-d^2/d\tau^2 + \omega^2)$:

$$\left(-\frac{d^2}{d\tau^2} + \omega^2\right) K(\tau, \tau') = \delta(\tau - \tau') \tag{2.30}$$

From (2.18), (2.29) and the definition (2.26) of $\Delta(\tau)$ we may identify K and Δ; in the case of the harmonic oscillator, we add a subscript F, where F stands for 'free', as the field theoretic generalization of the harmonic oscillator will be the free field:

$$K(\tau, \tau') = \Delta_F(\tau - \tau') \tag{2.31}$$

The solution of (2.30), given the periodicity condition (2.27), is unique. If τ lies in the interval $[0, \beta]$, we have

$$\Delta_F(\tau) = \frac{1}{2\omega}\left[(1 + n(\omega))e^{-\omega\tau} + n(\omega)e^{\omega\tau}\right] \tag{2.32}$$

where $n(\omega)$ is the Bose–Einstein distribution:

$$n(\omega) = \frac{1}{\exp(\beta|\omega|) - 1} \tag{2.33}$$

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An easy calculation allows one to check that (2.32), continued to negative values of τ thanks to the periodicity condition (2.27), does obey the differential equation (2.30), and that it is the unique solution of this equation with property (2.27): see exercise 2.3. One may also note that it is precisely the periodicity property of $\Delta_F(\tau)$ which allows one to perform the Gaussian integration in (2.29), as the periodicity of $q(\tau)$ is preserved in the change of integration variable (exercise 2.3).

2.2 Operator formalism

In the preceding section we already appealed to the operator formalism; however time was restricted to be pure imaginary. We now want to lift this restriction, as we are ultimately interested in real values of time. It is convenient to define the two-point functions $D^>(t,t')$ and $D^<(t,t')$:

$$D^>(t,t') = \langle \hat{q}(t)\hat{q}(t') \rangle_\beta \tag{2.34a}$$

$$D^<(t,t') = \langle \hat{q}(t')\hat{q}(t) \rangle_\beta = D^>(t',t) \tag{2.34b}$$

By inserting a complete set of eigenvectors of $\hat{H}$ one can express $D^>(t,t')$ as

$$D^>(t,t') = \frac{1}{Z(\beta)} \sum_{n,m} \mathrm{e}^{-\beta E_n} \mathrm{e}^{iE_n(t-t')} \times \mathrm{e}^{-iE_m(t-t')} | < n|\hat{q}(0)|m > |^2 \tag{2.35}$$

If the convergence in (2.35) is controlled by the exponentials, one sees that $D^>(t,t')$ is defined for

$$-\beta \leq \mathrm{Im}(t-t') \leq 0 \tag{2.36a}$$

while $D^<(t,t')$ is defined for

$$\beta \geq \mathrm{Im}(t-t') \geq 0 \tag{2.36b}$$

The functions $D^>$ and $D^<$ can be defined as distributions on the boundary of their domain of definition. We now use the fact that $\exp(-\beta\hat{H})$ is an evolution operator in imaginary time:

$$\mathrm{e}^{-\beta\hat{H}} \hat{q}(t) \mathrm{e}^{\beta\hat{H}} = \hat{q}(t + i\beta) \tag{2.37}$$

and we use the cyclicity of the trace to derive the Kubo–Martin–Schwinger (KMS) relation:

$$D^>(t,t') = D^<(t + i\beta, t') \tag{2.38}$$

It is useful to note that for τ lying in the interval $[0,\beta]$

$$\Delta(\tau) = D^>(-i\tau, 0) \tag{2.39}$$

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Then the periodicity condition (2.27) of $\Delta(\tau)$ follows from the KMS relation (2.38). Finally, for real values of t and t', we define the time-ordered propagator:

$$\begin{aligned} D(t,t') &= \langle T(\hat{q}(t)\hat{q}(t'))\rangle \\ &= \theta(t-t')D^{>}(t,t') + \theta(t'-t)D^{<}(t,t') \end{aligned} \tag{2.40}$$

The generalization of (2.40) to complex values of time will be dealt with in the following chapter.

2.3 The spectral function $\rho(k_0)$

The two-point Green function will appear in a variety of versions: imaginary time, real time, advanced, retarded. All these versions depend on a single function, which we call the spectral function $\rho(k_0)$. Because of translation invariance, $D^{>}(t,t')$ and $D^{<}(t,t')$ depend solely on the difference $t-t'$, and we shall use the shorthand notations $D^{>}(t) = D^{>}(t,0)$ and $D^{<}(t) = D^{<}(t,0)$. Let us define the Fourier transforms of $D^{>}(t)$ and $D^{<}(t)$:

$$D^{>}(k_0) = \int_{-\infty}^{\infty} \mathrm{d}t\, \mathrm{e}^{ik_0 t} D^{>}(t) \tag{2.41}$$

and

$$D^{<}(k_0) = \int_{-\infty}^{\infty} \mathrm{d}t\, \mathrm{e}^{ik_0 t} D^{<}(t) = \int_{-\infty}^{\infty} \mathrm{d}t\, \mathrm{e}^{ik_0 t} D^{>}(t - i\beta) \tag{2.42}$$

where the second equality follows from the KMS condition (2.38). Comparing (2.41) and (2.42) one deduces

$$D^{<}(k_0) = D^{>}(-k_0) = \mathrm{e}^{-\beta k_0} D^{>}(k_0) \tag{2.43}$$

The physical interpretation of (2.43), which is related to detailed balance, is examined in exercise 2.4. Moreover, one notes from the hermiticity of $\hat{q}(t)$ and translation invariance that $D^{>}(k_0)$ and $D^{<}(k_0)$ are real functions of k_0. The spectral function $\rho(k_0)$ is defined by

$$\rho(k_0) = D^{>}(k_0) - D^{<}(k_0) \tag{2.44}$$

and is thus the Fourier transform of the thermal average of the commutator $[\hat{q}(t), \hat{q}(0)]$. We can write from (2.43)

$$D^{>}(k_0) = (1 + f(k_0))\rho(k_0) \qquad D^{<}(k_0) = f(k_0)\rho(k_0) \tag{2.45}$$

with

$$f(k_0) = (\mathrm{e}^{\beta k_0} - 1)^{-1} \tag{2.46}$$

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An explicit expression for $\rho(k_0)$ can be obtained by using (2.35) and by computing $(D^>(k_0) - D^<(k_0))$, making use of (2.43):

$$\begin{aligned}\rho(k_0) &= \frac{2\pi}{Z(\beta)} \sum_{n,m} e^{-\beta E_n}(1 - e^{-\beta k_0}) \\ &\quad \times \delta(k_0 + E_n - E_m)| < n|\hat{q}(0)|m > |^2 \\ &= \frac{2\pi}{Z(\beta)} \sum_{n,m} e^{-\beta E_n}[\delta(k_0 + E_n - E_m) \\ &\quad - \delta(k_0 + E_m - E_n)]| < n|\hat{q}(0)|m > |^2 \end{aligned} \tag{2.47}$$

Equation (2.47) shows explicitly that

(i) $\rho(k_0)$ is an odd real function of k_0: $\rho(k_0) = -\rho(-k_0)$,
(ii) $\rho(k_0)$ obeys the positivity condition $\varepsilon(k_0)\rho(k_0) > 0$, where $\varepsilon(k_0)$ is the sign function.

Furthermore, $\rho(k_0)$ obeys a sum rule which can be deduced from the equal-time canonical commutation relation

$$\left[\hat{q}(t), \frac{d}{dt'}\hat{q}(t')\right]\Big|_{t'=t} = i \tag{2.48}$$

Indeed, we have

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{dk_0}{2\pi} k_0 e^{-ik_0 t} (D^>(k_0) - D^<(k_0)) \\ = i\frac{d}{dt}(D^>(t) - D^<(t))\end{aligned} \tag{2.49}$$

and, from the very definition of $D^>$ and $D^<$:

$$D^>(t) - D^<(t) = \langle[\hat{q}(t), \hat{q}(0)]\rangle_\beta \tag{2.50}$$

Taking the $t \to 0$ limit of the above equations yields the sum rule:

$$\int_{-\infty}^{+\infty} \frac{dk_0}{2\pi} k_0 \rho(k_0) = 1 \tag{2.51}$$

This sum rule can also be obtained from a simple manipulation of (2.47).

Finally, it will be interesting to write down the analogue of the propagator in quantum mechanics; it is easily found from the expression of the position operator in the case of the harmonic oscillator (with $m = 1$):

$$\hat{q}(t) = \frac{1}{\sqrt{2\omega}}\left(a e^{-i\omega t} + a^\dagger e^{i\omega t}\right) \tag{2.52}$$

(where $a^\dagger$ is the hermitian conjugate of a), and equation (2.50) which leads to

$$D^>(t) - D^<(t) = \frac{1}{2\omega}\langle[a, a^\dagger]e^{-i\omega t} + [a^\dagger, a]e^{i\omega t}\rangle_\beta \tag{2.53}$$

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Since $[a, a^\dagger] = 1$, one finds immediately the free (F) spectral function $\rho_F(k_0)$ by Fourier transformation

$$\rho_F(k_0) = 2\pi\varepsilon(k_0)\delta(k_0^2 - \omega^2) \tag{2.54}$$

It can easily be checked that this free spectral function obeys the positivity properties and the sum rule (2.51). It may also be worth noting that ρ_F is temperature-independent.

2.4 The Matsubara (or imaginary-time) propagator

We define the Fourier transform of the imaginary-time propagator $\Delta(\tau)$ (2.26) through

$$\Delta(i\omega_n) = \int_0^\beta d\tau\, e^{i\omega_n\tau}\Delta(\tau) \tag{2.55a}$$

the inverse formula being

$$\Delta(\tau) = T\sum_n e^{-i\omega_n\tau}\Delta(i\omega_n) \tag{2.55b}$$

Since $\Delta(\tau)$ is periodic ($\Delta(\tau - \beta) = \Delta(\tau)$), the Fourier transform is taken over a finite interval $[0, \beta]$, so that frequencies ω_n take discrete values:

$$\omega_n = \frac{2\pi n}{\beta} \tag{2.56}$$

which are called 'Matsubara frequencies'. If we choose τ in the interval $[0, \beta]$ we have $\Delta(\tau) = D^>(-i\tau)$ and, from (2.41):

$$\Delta(\tau) = \int \frac{dk_0}{2\pi} e^{-k_0\tau} D^>(k_0) \tag{2.57}$$

Using (2.55a) and the representation (2.45) of $D^>(k_0)$ in order to introduce the spectral function $\rho(k_0)$ leads to

$$\Delta(i\omega_n) = -\int_{-\infty}^{\infty} \frac{dk_0}{2\pi}\,\frac{\rho(k_0)}{i\omega_n - k_0} \tag{2.58}$$

In the free-field case, we use expression (2.54) for $\rho_F(k_0)$ and we get:

$$\Delta_F(i\omega_n) = \frac{1}{\omega_n^2 + \omega^2} \tag{2.59}$$

The inverse Fourier transform of (2.59) is, of course, $\Delta_F(\tau)$ given in (2.32). The Matsubara (full) propagator (2.58) is defined for discrete values (2.56) of the frequencies only and its analytic continuation to arbitrary values of the frequencies is not unique. However, from (2.58) one can define a unique analytic continuation provided that one requires

(i) $|\Delta(z)| \to 0$ if $|z| \to \infty$;
(ii) $\Delta(z)$ is analytic outside the real axis.

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Then the analytic continuation is provided by

$$\Delta(z) = -\int_{-\infty}^{\infty} \frac{dk_0}{2\pi} \frac{\rho(k_0)}{z - k_0} \tag{2.60}$$

This is in fact the continuation we are interested in, since for real values of z, or, more exactly, for $z = k_0 \pm i\eta$, $\eta \to 0^+$, with k_0 real, we obtain the retarded and advanced propagators which occur in linear response theory (see chapter 6) $D_R(t)$ and $D_A(t)$:

$$D_R(t) = \langle \theta(t)[\hat{q}(t), \hat{q}(0)] \rangle_\beta \tag{2.61}$$

$$D_A(t) = -\langle \theta(-t)[\hat{q}(t), \hat{q}(0)] \rangle_\beta \tag{2.62}$$

Indeed, from the usual representation of the θ-function:

$$\theta(t) = i \int_{-\infty}^{\infty} \frac{dk_0'}{2\pi} \frac{e^{-ik_0' t}}{k_0' + i\eta} \tag{2.63}$$

one finds, for example, the Fourier transform $D_R(k_0)$ of $D_R(t)$:

$$D_R(k_0) = i \int_{-\infty}^{\infty} \frac{dk_0'}{2\pi} \frac{\rho(k_0')}{k_0 - k_0' + i\eta} \tag{2.64}$$

so that:

$$D_R(k_0) = -i\Delta(k_0 + i\eta) \qquad D_A(k_0) = i\Delta(k_0 - i\eta) \tag{2.65}$$

The function $D_R(k_0)$ is analytic in the upper half complex k_0 plane. Moreover, we note that the *free* retarded and advanced Green functions are temperature-independent because the free spectral function (2.54) is itself temperature-independent. This is not surprising: as will be explained in chapter 6, the physical content of these Green functions is that they give the energies and the lifetimes of the elementary excitations. In the free case we simply have $k_0 = \pm\omega$ and the excitations are of course stable.

2.5 The time-ordered propagator

We now wish to establish the expression of the thermal propagator in real time. The Fourier transform $D(k_0)$ of the propagator (2.40) reads

$$D(k_0) = \int dt\, e^{ik_0 t} \Big(\theta(t) D^>(t) + \theta(-t) D^<(t) \Big) \tag{2.66}$$

From the representation (2.63) of the θ-function one derives

$$D(k_0) = i \int \frac{dk_0'}{2\pi} \frac{\rho(k_0')}{k_0 - k_0' + i\eta} + f(k_0)\rho(k_0) \tag{2.67}$$

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From this expression it is clear that $D(k_0)$ is *not* the analytic continuation toward real frequencies of the Matsubara propagator. Indeed, we may define a continuation D' which depends on the sign of k_0:

$$D'(k_0) = -i\Delta(k_0 + i\eta k_0) \tag{2.68}$$

and we have

$$D(k_0) = (1 + n(k_0))D'(k_0) + n(k_0)D'^*(k_0) \tag{2.69}$$

where

$$n(k_0) = (e^{\beta|k_0|} - 1)^{-1} \tag{2.70}$$

It is now easy to find the free propagator by substituting the expression (2.54) of the free spectral function in (2.67):

$$D_F(k_0) = \frac{i}{k_0^2 - \omega^2 + i\eta} + 2\pi n(k_0)\delta(k_0^2 - \omega^2) \tag{2.71}$$

There are two important points to be noted in the preceding formula. First, one sees that (2.71) allows one to separate a zero-temperature part and a thermal part, which vanishes in the $T = 0$ limit. Second, one has to note that it is the absolute value $|k_0|$ which is featured in (2.70), while k_0 appears in (2.46). We could hope that (2.71), suitably generalized to take the spatial degrees of freedom into account, would give the time-ordered propagator at non-zero T. Unfortunately, as we shall see in the next chapter, the full story is more complicated (even in ordinary quantum statistical mechanics!), and one has to double the field degrees of freedom in order to obtain the correct perturbative expansion.

2.6 Frequency sums

In calculations using the imaginary-time formalism, one often encounters sums over Matsubara frequencies. There exist standard techniques to handle these frequency sums. In the present section, we shall explain one of these techniques, which relies on the representation (2.55) of the Matsubara propagator and the explicit expression (2.32) of $\Delta_F(\tau)$. Another useful technique, using contour integration, will be explained in the following chapter.

Our first example concerns the calculation of the partition function of the harmonic oscillator. Of course, this partition function is most easily obtained by using definition (2.11) and a complete set of states of the Hamiltonian. Here we just want to gain some familiarity with path integrals in thermal field theory, and we try to derive $Z(\beta)$ from (2.15). The functional

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integral is of the Gaussian type and we start from the basic formula:

$$\int_{-\infty}^{\infty} \prod_{i=1}^{i=N} \mathrm{d}\varphi_i \exp\left[-\frac{1}{2}\sum_{i,j} \varphi_i A_{ij}^{-1} \varphi_j\right]$$
$$= (2\pi)^{N/2}(\det A)^{1/2} = (2\pi)^{N/2} \exp\left[\frac{1}{2}\mathrm{Tr}\ln A\right] \tag{2.72}$$

Within an additive constant, which we can drop because it does not affect thermodynamics, we obtain

$$\ln Z(\beta) = \frac{1}{2}\mathrm{Tr}\ln K = \frac{1}{2}\mathrm{Tr}\ln \Delta_F \tag{2.73}$$

where K is the operator defined in (2.30). The trace is computed by going to Fourier space, where Δ_F is diagonal, and by using representation (2.59)

$$\ln Z(\beta) = -\frac{1}{2}\sum_{n=-\infty}^{n=+\infty} \ln(\omega^2 + \omega_n^2) \tag{2.74}$$

One notes that the sum over n is divergent: this unfortunate feature stems from our careless handling of the integration measure $\mathscr{D}q(\tau)$. A more rigorous treatment (see exercise 2.2), using the proper definition (2.8) of $\mathscr{D}q(\tau)$, gives a perfectly finite result. In order to handle (2.74), we differentiate with respect to ω:

$$\frac{1}{2\omega}\frac{\mathrm{d}\ln Z(\beta)}{\mathrm{d}\omega} = -\frac{1}{2}\sum_{n=-\infty}^{n=+\infty} \frac{1}{(\omega^2 + \omega_n^2)}$$
$$= -\frac{\beta}{2}\Delta_F(\tau = 0) = -\frac{\beta}{4\omega}(1 + 2n(\omega)) \tag{2.75}$$

Then, by integration with respect to ω, we derive the free energy:

$$-\frac{1}{\beta}\ln Z(\beta) = \frac{1}{2}\omega + \frac{1}{\beta}\ln\left(1 - \mathrm{e}^{-\beta\omega}\right) + \text{constant} \tag{2.76}$$

where the constant is divergent, but independent of β and ω; (2.76), without the additive constant, is of course the standard result which can be found in any textbook on statistical mechanics in the part which is devoted to blackbody radiation.

Another typical frequency sum, which we shall encounter in the calculation of loop diagrams, is

$$S(i\omega_m) = T\sum_n \Delta(i(\omega_m - \omega_n))\Delta'(i\omega_n) \tag{2.77}$$

where the oscillator frequencies of Δ and Δ' are ω and ω' respectively. Using the Fourier representation (2.55) of $\Delta(i\omega_n)$ and the relation

$$T\sum_n \mathrm{e}^{i\omega_n\tau} = \sum_p \delta(\tau - p\beta) \tag{2.78}$$

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where $p = 0, \pm 1, \pm 2, \ldots$, yields

$$S(i\omega_n) = \int_0^\beta d\tau\, e^{i\omega_n \tau} \Delta(\tau)\Delta'(\tau) \tag{2.79}$$

We now express $\Delta(\tau)$ and $\Delta'(\tau)$ in terms of their spectral functions $\rho(k_0)$ and $\rho'(k_0')$ (see (2.45) and (2.57)), integrate over τ remembering that $\exp(i\omega_m \beta) = 1$, and make use of simple identities involving the Bose–Einstein distribution to derive

$$\begin{aligned} S(i\omega_m) = & -\int_{-\infty}^{+\infty} \frac{dk_0}{2\pi} \frac{dk_0'}{2\pi} \\ & \times (1 + f(k_0) + f(k_0')) \frac{\rho(k_0)\rho'(k_0')}{i\omega_m - k_0 - k_0'} \end{aligned} \tag{2.80}$$

As explained in section 2.4, this expression can be analytically continued to values of ω_m which are not Matsubara frequencies, and in particular to real values of the energy: $i\omega_m \to q_0$, but of course only after the frequency sum over ω_n has been performed. Let us specialize this formula to the case of the 'free propagators' (2.59); we set $k_0 = s\omega$, $s = \pm 1$, and note that

$$f(\omega) = n(\omega) \qquad f(-\omega) = -(1 + n(\omega)) \tag{2.81}$$

to write the final result as

$$\begin{aligned} S(q_0) = & -\frac{1}{4\omega\omega'} \Big[(1 + n(\omega) + n(\omega')) \\ & \times \Big(\frac{1}{q_0 - \omega - \omega'} - \frac{1}{q_0 + \omega + \omega'} \Big) + (n(\omega) - n(\omega')) \\ & \times \Big(\frac{1}{q_0 + \omega - \omega'} - \frac{1}{q_0 - \omega + \omega'} \Big) \Big] \end{aligned} \tag{2.82}$$

This formula will lie at the basis of the one-loop calculations of the following chapters. We shall also need the imaginary part of $S(q_0 + i\eta)$, $\eta \to 0^+$; for arbitrary spectral functions we obtain from (2.80)

$$\begin{aligned} \mathrm{Disc}\, S(q_0) &= S(q_0 + i\eta) - S(q_0 - i\eta) = 2i\, \mathrm{Im}\, S(q_0 + i\eta) \\ &= 2i\pi \int_{-\infty}^{+\infty} \frac{dk_0}{2\pi} \frac{dk_0'}{2\pi} (1 + f(k_0) + f(k_0')) \\ &\quad \times \rho(k_0)\rho'(k_0')\delta(q_0 - k_0 - k_0') \end{aligned} \tag{2.83}$$

An equivalent formula is

$$\begin{aligned} \mathrm{Disc}\, S(q_0) = & 2i\pi (e^{\beta q_0} - 1) \int_{-\infty}^{+\infty} \frac{dk_0}{2\pi} \frac{dk_0'}{2\pi} \\ & \times f(k_0) f(k_0') \rho(k_0)\rho'(k_0')\delta(q_0 - k_0 - k_0') \end{aligned} \tag{2.84}$$

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In the case of free propagators, (2.83) becomes

$$\begin{aligned}\operatorname{Disc} S(q_0) = \frac{2i\pi}{4\omega\omega'}\Big[&(1+n(\omega)+n(\omega'))(\delta(q_0-\omega-\omega') \\ &-\delta(q_0+\omega+\omega')) + (n(\omega)-n(\omega')) \\ &\times(\delta(q_0+\omega-\omega')-\delta(q_0-\omega+\omega'))\Big] \end{aligned} \tag{2.85}$$

References and further reading

There are many good accounts of path integrals in textbooks: see, for example, Feynman (1972), chap. 3; Schulman (1981); Rivers (1987); Negele and Orland (1988), chap. 1; Parisi (1988), chap. 13; Brown (1992), chap. 1. The integration measure in path integrals is examined from a mathematical point of view by Glimm and Jaffe (1987) and the extension of (2.5) to unbounded operators by Glimm and Jaffe (1987) and Schulman (1981). The analytical continuation from Matsubara frequencies to real values of the energy was first discussed by Baym and Mermin (1961); see also Fetter and Walecka (1971), chap. 9. Our presentation of sections 2.2–2.5 has been inspired by Dolan and Jackiw (1974), who were the first to derive the propagator (2.71), and Blaizot (1992).

Exercises

2.1 Obtain the Hamiltonian form of the path integral by writing $\exp(-\varepsilon\hat{H}) \simeq 1-\varepsilon\hat{H}$ and leaving the p-integration in (2.7) undone:

$$\begin{aligned}F(q',-i\tau';q,-i\tau) = \lim_{\varepsilon\to 0}\int\prod_{l=1}^{n}\mathrm{d}q_l\prod_{l=0}^{n}\Big[\frac{\mathrm{d}p_l}{2\pi} \\ \times\exp(ip_l(q_{l+1}-q_l))\exp(-\varepsilon H(p_l,q_l))\Big]\end{aligned}$$

which is written formally as ($\dot{q} = \mathrm{d}q/\mathrm{d}\tau$)

$$\int \mathscr{D}p\mathscr{D}q\exp\Big(\int_{\tau}^{\tau'}\mathrm{d}\tau\Big[ip\dot{q}-H(p,q)\Big]\Big)$$

2.2 We wish to compute the harmonic oscillator partition function from its path integral representation (2.15). We begin with the evaluation of the Euclidean action:

$$S[q(\tau)] = \frac{1}{2}\int_0^{\beta}\mathrm{d}\tau\Big(\dot{q}^2(\tau)+\omega^2q^2(\tau)\Big)\mathrm{d}\tau$$

with $q(0) = q(\beta) = q$ (the calculation may be easily generalized to boundary conditions $q(0) = q'$, $q(\beta) = q''$). Let $q_{\mathrm{cl}}(\tau)$ be a solution of the

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equation of motion $(d^2q_{cl}/d\tau^2 - \omega^2 q_{cl}) = 0$ with $q_{cl}(0) = q_{cl}(\beta) = q$; we write

$$q(\tau) = q_{cl}(\tau) + h(\tau)$$

Use the stationarity of the action ($[\delta S/\delta q]_{q=q_{cl}} = 0$) and the equation of motion to obtain

$$\begin{aligned} S[q(\tau)] &= \frac{1}{2} q[\dot{q}(\beta) - \dot{q}(0)] + S[h(\tau)] \\ &= \frac{q^2\omega(\cosh \beta\omega - 1)}{\sinh \beta\omega} + S[h(\tau)] \end{aligned}$$

In order to evaluate

$$Z' = \int \mathscr{D}h \exp(-S[h(\tau)])$$

we discretize the h-integration. The determinant to be evaluated takes the form

$$D_n = \begin{pmatrix} 1 & -a & 0 & \dots & \\ -a & 1 & -a & 0 & \dots \\ 0 & \ddots & & & \\ \vdots & & & & \end{pmatrix} \qquad |a| < \frac{1}{2}$$

Show that $D_n = D_{n-1} - a^2 D_{n-2}$ and solve this recursion relation. Taking the limit $n \to \infty$ yields

$$Z' = \left(\frac{\omega}{2\pi \sinh \beta\omega}\right)^{1/2}$$

Perform the q-integration and recover (2.76) with the constant $= 0$.

2.3 (a) Use the change of variable

$$q'(\tau) = q(\tau) - \int_0^\beta K(\tau, \tau') j(\tau')\, d\tau'$$

in the first two lines of (2.29) in order to obtain the last line of (2.29). Show that this change of variable preserves, as it should, the periodicity of $q(\tau)$.

(b) Show that (2.32) is the unique solution of (2.30) with the condition (2.27).

2.4 Assume that a harmonic oscillator in a thermal bath is submitted to a weak external force $j(t)$, so that the extra Hamiltonian is $j(t)\hat{q}(t)$. The transition probability $n \to m$ is given by Fermi's golden rule; it is proportional to

$$w_{n\to m} = 2\pi\delta(k_0 + E_n - E_m) | < n|\hat{q}|m > |^2$$

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Show that (2.47) can be written as

$$\rho(k_0) = \sum_{n,m} (P_n - P_m) w_{n \to m}$$

where P_n is the probability of occupation of level n. Show that for $k_0 > 0$, $D^>$ gives the rate of energy absorption by the oscillator, while $D^<$ gives the rate of energy emission.

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